Supporting information

Caesarean section and obesity in young adult offspring: update of a systematic review with meta-analysis

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1 METHODS

Search strategies

The following search strategies were used:

[1] PubMed: (Delivery, obstetric [MeSH] OR Cesarean Section [MeSH] OR Cesarean [TI/AB]) AND (Obesity, Abdominal [MeSH] OR Obesity, Metabolically Benign [MeSH] OR Obesity, Morbid [MeSH] OR Abdominal circumference OR Body Weights and Measures [MeSH] OR Waist Circumference [MeSH] OR Waist-Hip Ratio [MeSH] OR Body Mass Index [MeSH] OR Body Fat Distribution [MeSH] OR Waist-Height Ratio [MeSH] OR Height Weight Ratio [MeSH] OR Abdominal Fat [MeSH] OR Fat Body [MeSH]).

[2] EMBASE: ('Delivery, obstetric'/de OR 'Cesarean Section'/exp OR Cesarean*:ti OR Cesarean:ab) AND ('abdominal obesity'/exp OR 'metabolically benign obesity'/exp OR 'adolescent obesity'/exp OR 'morbid obesity'/exp OR 'abdominal circumference'/exp OR 'body mass'/exp OR 'body weight control'/exp OR 'hip circumference'/exp OR 'waist circumference'/exp OR 'waist hip ratio'/exp OR 'waist to height ratio'/exp OR 'weight'/exp OR 'weight height ratio'/exp OR 'abdominal fat'/exp OR 'body fat'/exp OR 'body fat distribution'/exp).

Study selection

Discrepancies in the screening were resolved by consensus with a third person [CC or SC]. The full texts of relevant abstracts were appraised for inclusion [YG and BQ]. Any disagreement over eligibility of a study was referred to a meeting with a third person [CC or SC].

Statistical analysis

We reported risk ratios as a common measure of the association. To harmonize the measure across cohort studies, we transformed odds ratios into risk ratios when obesity was common. This was the case for the study of Mamun et al. [1] in which the prevalence of obesity was 12.5% but not for the study of Hansen et al. [2] in which it was 8%. Furthermore, we interpreted the prevalence ratios reported in Goldani et al.,[3] Barros et al.,[4] and Svensson et al.[5] as risk ratios because the recalled or self-reported exposure occurred at birth for all participants. Finally, we approximated the rate ratio reported in Sogunle et al.[6] as a risk ratio by assuming a similar length of follow-up for exposed and non-exposed participants because the outcome was measured in a narrow window of ages (21-24 years).

Quality assessment

To adjust for bias, we used a quality assessment instrument (Table S1) created by Sutharsan et al.[7] based on bias criteria for observational studies.[8] The quality score was calculated by dividing the total points received by the maximum number of 18 points.

Figure S1: Sutharsan quality assessment instrument [7]

A. Design-specific bias		
Q1	*Was the data completely prospectively collected?*	0 = No/others; 1 = Yes
B. Selection Bias		
Q2	*Were consecutive subjects recruited into the study?*	0 = No /others 1= Yes
Q3	*Were the number of participants lost at each stage or protocol deviations, acceptable (based on reason for dropout) or <20%? (justify definition of "acceptable dropouts")*	0 = No or not reported 1 = Yes
Q4	*Were the non-exposed cohort drawn from the same population and in the same way as the exposed cohort?*	0 = No or no description 1 = Drawn from a different source 2 = Yes
Q5	*Were the eligibility criteria clearly specified and applied in the same way in the exposed and unexposed?*	0 = No or no description 0.5 = In part 1 = Yes
C. Confounding		
Q6a	*Was any attempt made to adjust for potential confounders ?* (Birth weight, Maternal pre-preg. BMI, Gestational weight gain, Maternal complications, Gestation, SES, parity, age, race)	0 = None 1 = 2-4 confounders 2 = ≥5 confounders
Q6b	If adjusted for maternal pre-preg. BMI	4 = Yes
Q6c	If adjusted for maternal complications or gestational weight gain	2 = Yes
D. Information Bias		
Q7	*Was timing of outcome assessment and duration of follow-up adequate for outcomes to occur?*	No - 0 = <1 year Yes - 1 ≥ 1 year
Q8	*How accurately was outcome measured?*	0 = self-reported 0.5 = reported in chart/Recorded 0.75 = single (one time point) measured by investigators 1 = multiple measured (child/adolescent/adult) by investigators
E. Statistical methods		
Q9	*Was the analysis reported on the entire data?*	0 = Not clearly documented 1 = Yes
Q10	*Was missing data analysed using imputation methods?*	0 = no 1 = yes or missing data < 20%

2 RESULTS

Figure S2: Flow diagram of the selection of articles for the meta-analysis

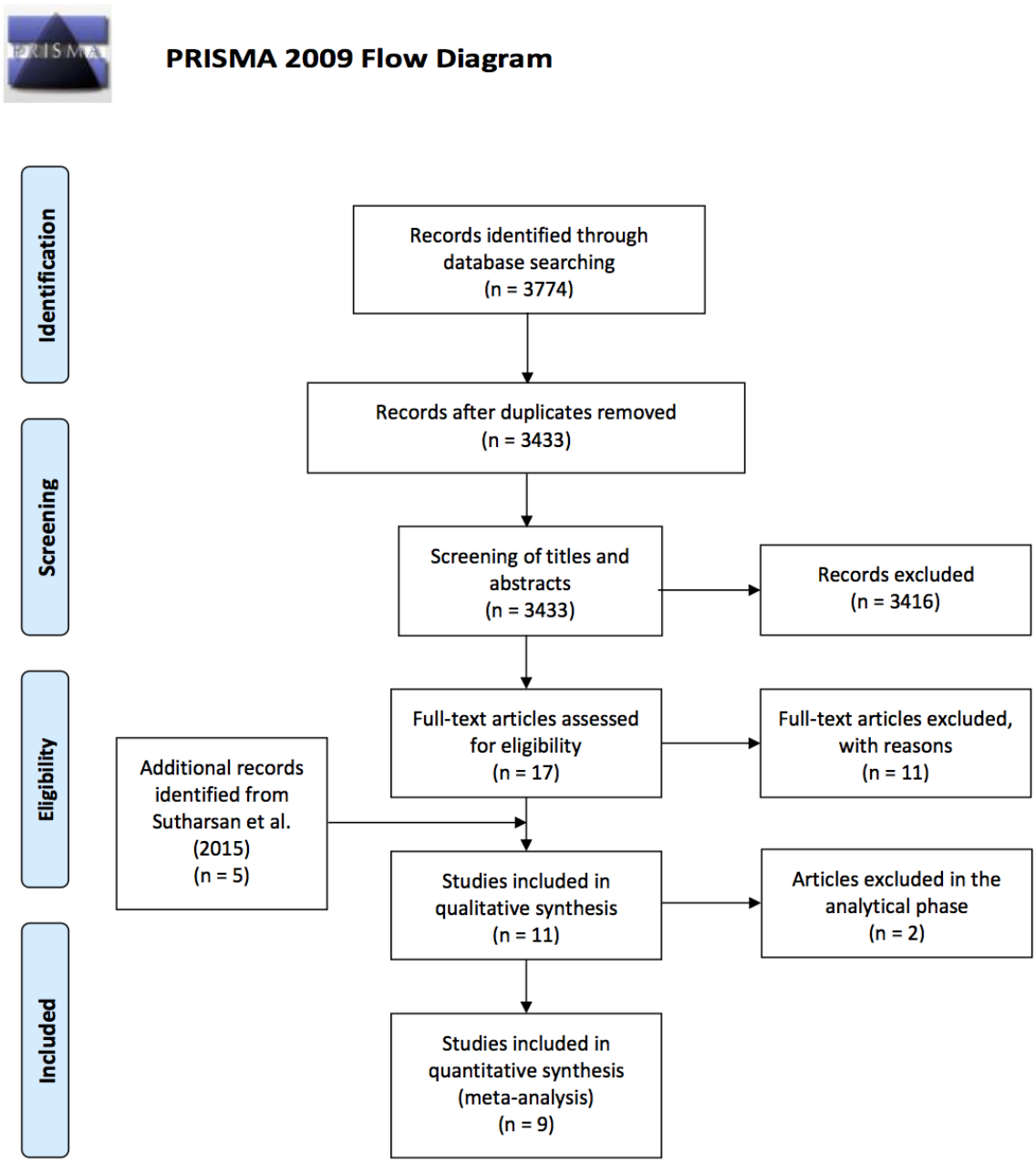
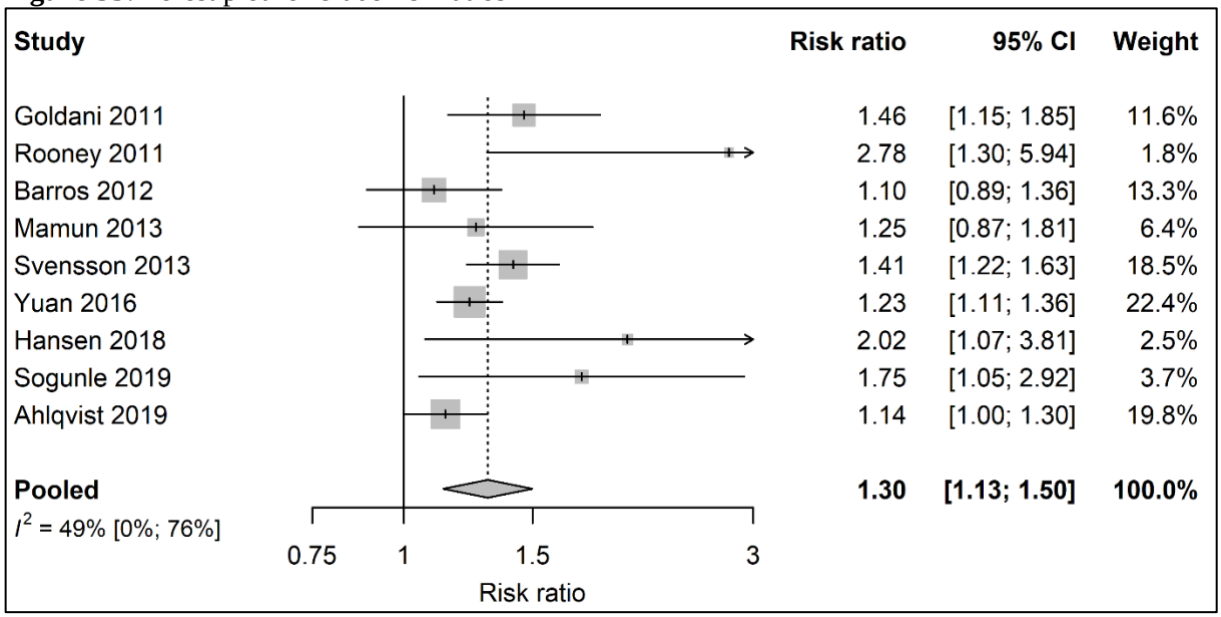
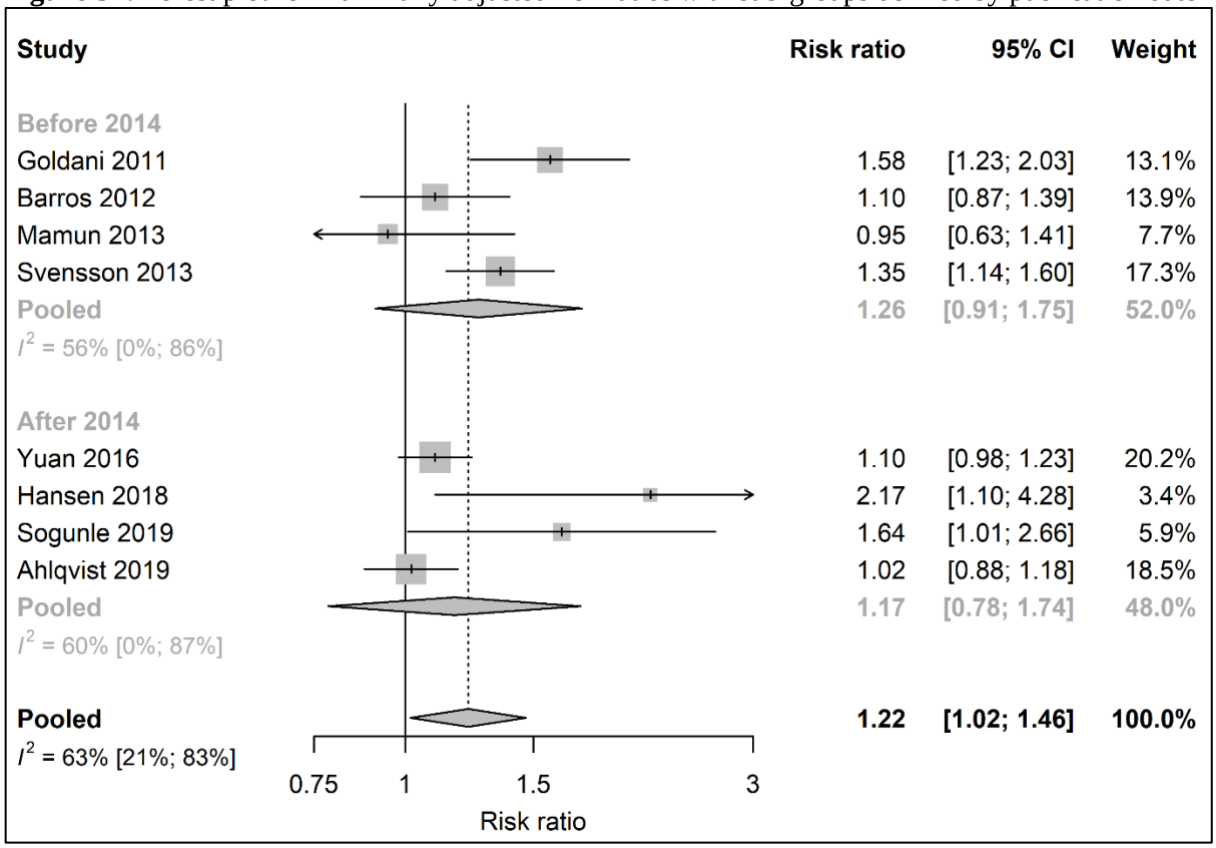


Figure S3: Forest plot for crude risk ratios



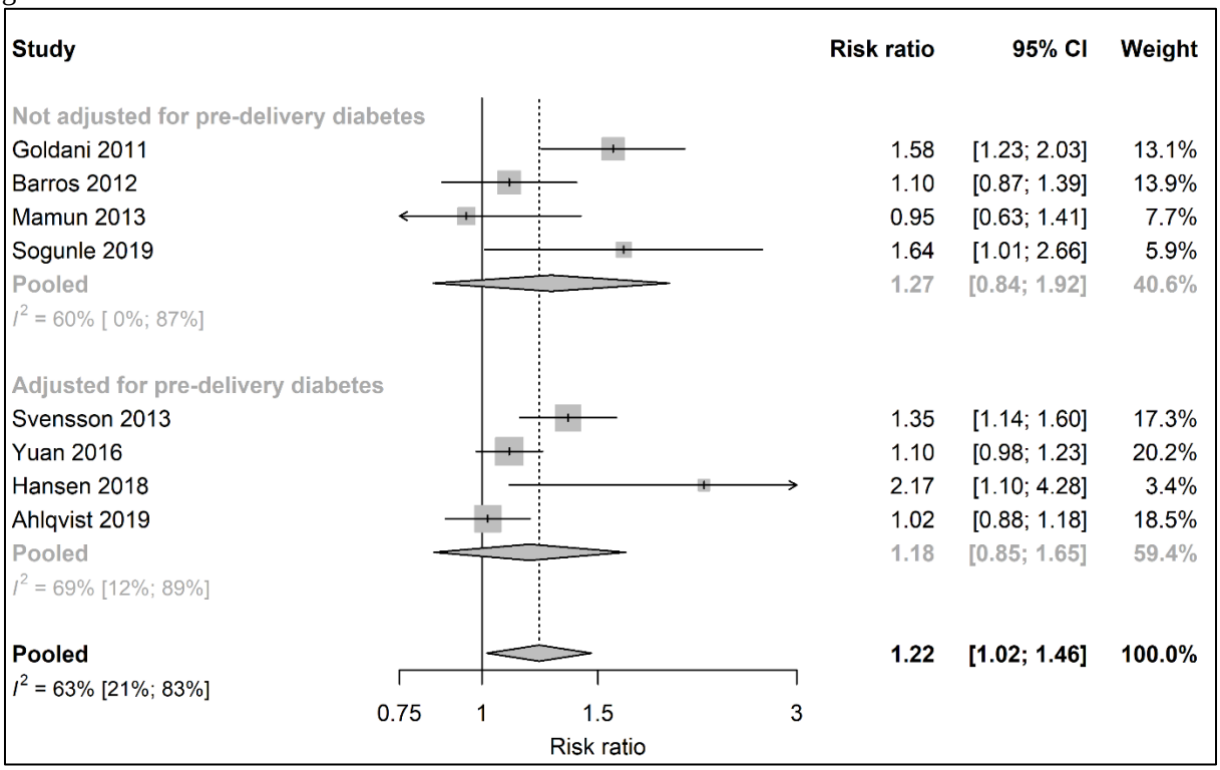
Pooled estimates are represented by a diamond. The size of the grey square for each study is related to the amount of variance that a study contributes to the meta-analysis (column "weight").

Figure S4: Forest plot for maximally adjusted risk ratios with sub-groups defined by publication date



Pooled estimates are represented by a diamond. The size of the grey square for each study is related to the amount of variance that a study contributes to the meta-analysis (column "weight").

Figure S5: Forest plot for maximally adjusted risk ratios with sub-groups defined by adjustment with gestational diabetes mellitus



Pooled estimates are represented by a diamond. The size of the grey square for each study is related to the amount of variance that a study contributes to the meta-analysis (column "weight").

Figure S6: Funnel plot for publication bias

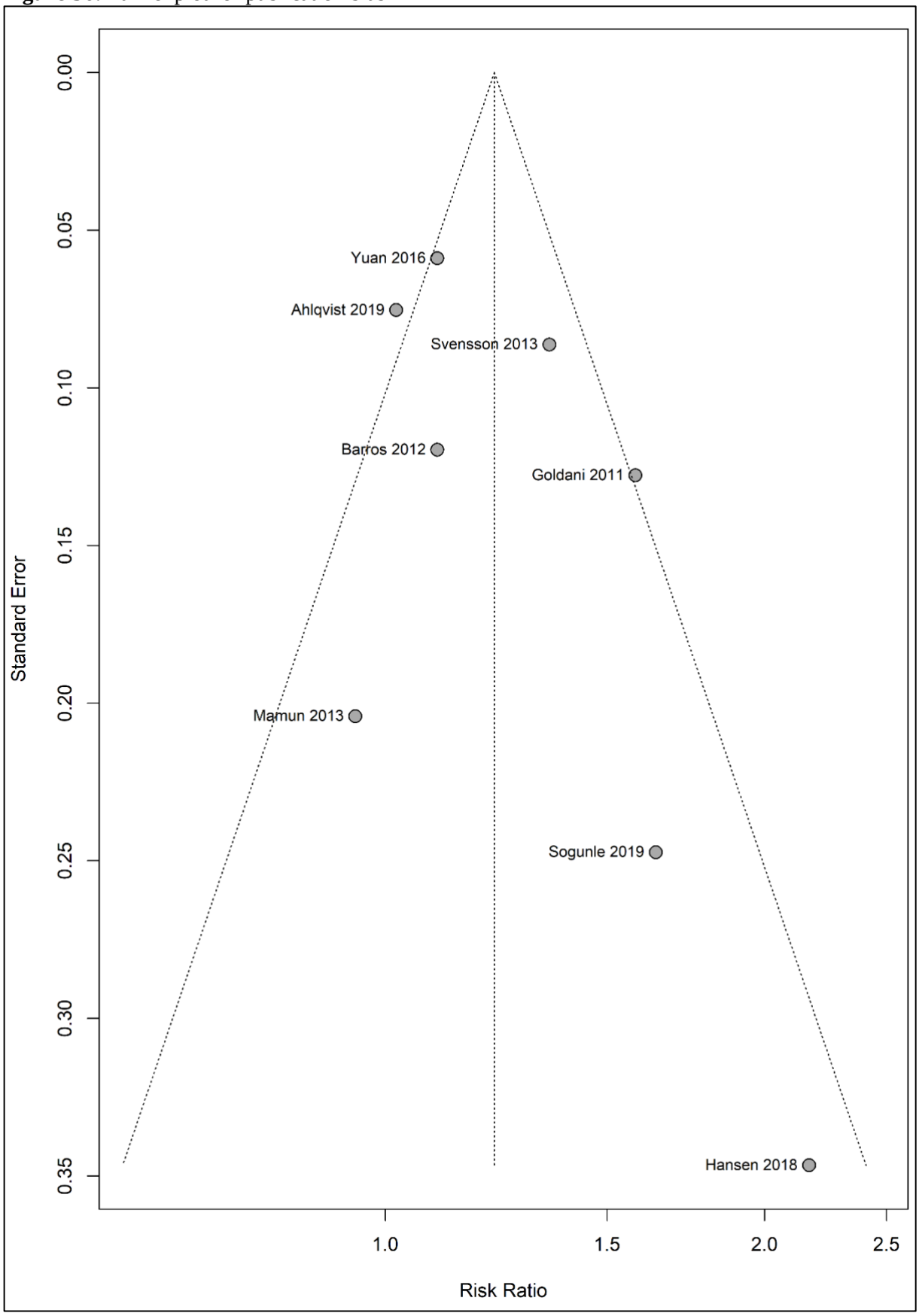


Table S1: Characteristics of longitudinal studies not included in the meta-analysis

First author, year	Sample, year of exposure, country	Sample size	Mean age or age range	Out-come	Definition of obesity	Measure of association	Crude estimates	Fully adjusted estimates	Adjusted for maternal pre-pregnancy BMI	Factors adjusted for	Quality Score
Bernardi, 2015[9]	Population based, 1978–1979, Brazil	2,063 M = 995 F = 1,068	23 ± 0.7 years	BMI	not dichoto-mized	B	0.018 (0.010, 0.025) n.a.	0.020 (0.013, 0.028) n.a.	No	Maternal factors: schooling at birth, age, smoking during pregnancy Child factors: birth weight, sex, parity	0.54
Barros, 2017[10]	Population based, 1982-2013, Brazil 1982 cohort	3,607 M = 1,754 F = 1,853	30 years	BMI z-score	not dichoto-mized	B	0.13 (0.06, 0.21) 0.12 (0.03, 0.21) 0.14 (0.03, 0.26)	0.09 (0.01, 0.17) 0.04 (-0.06, 0.14) 0.15 (0.03, 0.28)	Yes	Maternal factors: education, skin colour, pre-gestational BMI, parity, age at birth and smoking during pregnancy. Family factors: SEP (socioeconomic position), financing of delivery Child factors: birth weight ditto ditto	0.72
	1993 cohort	3,961 M = 1,965 F = 1,996	18 years	BMI z-score	not dichoto-mized	B	0.17 (0.09, 0.25) 0.27 (0.15, 0.38) 0.07 (−0.04, 0.19)	0.06 (-0.02, 0.14) 0.11 (-0.01, 0.23) 0.01 (-0.11, 0.13)	Yes	Maternal factors: education, skin colour, pre-gestational BMI, parity, age at birth and smoking during pregnancy. Family factors: SEP (socioeconomic position), financing of delivery Child factors: birth weight and length ditto ditto	

B: Beta (multiple linear regression), BMI: body mass index, CS: caesarean section, F: female, M: male, n.a.: not available, ob: obesity, RR: relative risk

3 DISCUSSION

Table S2: Comparison with three previous meta-analyses

	Li et al. 2013[11]	**Darmasseelane et al. 2014**[12]	**Sutharsan et al. 2015**[7]	**Present study (Update of Sutharsan 2015)**
Type of studies included	Longitudinal studies Case-control studies	Long term follow-up studies	Longitudinal studies	Longitudinal studies
Number of studies included in analysis (regarding effect on young adulthood)	3	11	5	4 plus 5 (crude analysis) respectively 4 (adjusted analysis) from Sutharsan
Age cut-off for young adulthood	≥19 years	≥18	≥18	≥18
Number of participants	6,807	≈35,000	30,231	143,869 (crude) 143,416 (adjusted)
Consideration for quality of studies	- Newcastle-Ottawa scale - Subgroup analysis not conducted for adulthood	- Newcastle-Ottawa scale - Subgroup analysis by high-, medium- and low-quality studies	Assessment instrument developed by Sutharsan et al., based on bias criteria for observational studies	Studies assessed with the Sutharsan quality score
Effects estimate (RR/OR)	OR	OR	Relative risks (OR transformed in relative risks)	Risk ratios (OR transformed in relative risk; PR and rate ratio interpreted as risk ratios)
Type of model applied	Random-effects model MA, with inverse variance method	Fixed and random-effects model MA, with Mantel-Haenszel method	Meta-regression of relative risks adjusted for synthetic bias variance using on a ranking based on studies' quality score	Random-effects model MA, with DerSimonian-Laird method and Hartung-Knapp adjustment
Results Unadjusted estimates for adulthood obesity	(no)	1.22 (95% CI 1.05-1.42); $I^2 = 22\%$	(no)	1.30 (95% CI 1.13-1.50); $I^2 = 49.0\%$
Adjusted estimates for adulthood obesity	1.50 (95% CI 1.02-2.20); $I^2 = 74\%$	(no)	1.28 (95% CI 1.02-1.34)	1.22 (95% CI 1.02-1.46); $I^2 = 63.0\%$
Consideration for confounders	Estimation of adjusted risk ratios	Not performed	Confounding can count up to half the score points	- Estimation of adjusted risk ratios - Sub-group analysis by confounding factor

Publication bias	Methods: Funnel plot with Egger's test Results: analysis not conducted (not enough studies)	Methods: Funnel plot with Egger's test Results: Asymmetry of funnel plot, Egger's test NS	Methods: DOI and funnel plots with Egger's test Results: analysis not conducted (not enough studies)	Methods: Funnel plot Results: no asymmetry pattern

CI: confidence intervals, MA: meta-analysis, NS: not significant, OR: odds ratio, PR: prevalence ratio; RR, risk ratio

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